

Lec8

CSS Grid Layout

CSS Grid Layout is a two-dimensional layout system in CSS that allows you to arrange elements into **rows and columns**. It is used to create complex webpage layouts easily and efficiently.

Why use CSS Grid?

Creates responsive layouts.

Controls both rows and columns simultaneously.

Reduces the need for floats and positioning.

Makes alignment and spacing easier.

Example (HTML)

```
div class="container">
```

```
<div class="item">1</div>
```

```
<div class="item">2</div>
```

```
<div class="item">3</div>
```

```
<div class="item">4</div>
```

```
</div>
```

Example (CSS)

```
.container {  
  
display: grid;  
  
grid-template-columns: 100px 100px;  
grid-template-rows: 100px 100px;  
  
gap: 10px;  
  
}
```

```
.item {  
  
background-color: lightblue;  
  
border: 1px solid black;  
  
}
```

Grid Properties

display: grid

Makes an element a grid container.

```
.container {  
  display: grid;  
}
```

grid-template-columns

Defines the number and size of columns.

grid-template-columns: 100px 200px 100px;

Using fractions:

grid-template-columns: 1fr 2fr 1fr;

grid-template-rows

Defines row sizes.

```
grid-template-rows: 100px 150px;
```

gap

Adds space between rows and columns.

```
gap: 20px;
```

or

```
row-gap: 20px;
```

```
column-gap: 10px;
```

grid-column

Makes an item span multiple columns.

```
.item1 {  
  grid-column: 1 / 3;  
}
```

grid-row

Makes an item span multiple rows.

```
.item1 {  
  grid-row: 1 / 3;  
}
```


justify-items

Aligns items horizontally inside grid cells.

```
justify-items: start;  
justify-items: center;  
justify-items: end;  
justify-items: stretch;
```

align-items

Aligns items vertically inside grid cells.

```
align-items: start;  
align-items: center;  
align-items: end;  
align-items: stretch;
```

justify-content

Aligns the entire grid horizontally.

```
justify-content: center;  
justify-content: start;  
justify-content: end;  
justify-content: space-between;  
space-around;  
space-evenly;
```

align-content

Aligns the entire grid vertically.

```
align-content: center;  
align-content: start;  
align-content: end;  
align-content: space-between;
```

Flexbox vs Grid

Flexbox	Grid
One-dimensional	Two-dimensional
Row OR Column	Row AND Column
Best for navigation bars	Best for page layouts
Simpler	More powerful

Responsive Grid Design

Responsive Grid Design is a web design technique where a grid layout automatically adjusts its rows and columns according to different screen sizes (mobile, tablet, laptop, desktop).

The goal is to make a website look good on every device without creating separate layouts for each screen size.

Example (HTML)

```
<div class="container">
```

```
<div class="card">Card 1</div>
```

```
<div class="card">Card 2</div>
```

```
<div class="card">Card 3</div>
```

```
<div class="card">Card 4</div>
```

```
</div>
```

Example (CSS)

```
.container{  
  display:grid;  
  grid-template-columns:  
    repeat(auto-fit,minmax(200px,1fr));  
  gap:20px;  
}
```

```
.card{  
  background:lightblue;  
  padding:20px;  
}
```

Task

Create a responsive image gallery using Grid